

NAIMAKKA, Sweden

WMO: 020550

Lat: 68.6762N

Lon: 21.5229E

Elev: 402

StdP: 96.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-31.0	-37.2	0.1	-33.4	-34.2	0.2	-30.8	11.9	-2.3	10.4	-3.2	1.2	260	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	22.5	14.9	20.2	13.9	18.0	12.8	16.1	20.5	14.9	18.7	13.7	17.0	3.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	10.7	17.6	13.2	9.9	16.3	12.0	9.2	15.3	46.5	20.8	43.0	18.5	39.8	16.9	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.2	7.0	-38.7	24.8	2.9	1.8	-40.8	26.0	-42.5	27.1	-44.1	28.1	-46.2	29.3
			-38.3	17.3	2.5	1.3	-40.1	18.3	-41.6	19.0	-43.0	19.7	-44.9	20.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-14.8	-14.5	-9.3	-3.6	3.0	8.3	12.4	10.2	5.5	-1.7	-8.8	-11.2
	DBStd	10.97	8.45	8.34	6.31	4.65	3.87	3.31	3.22	3.27	3.48	4.90	6.84	8.22
	HDD10.0	4523	769	684	597	408	218	72	12	38	139	363	565	657
	HDD18.3	7409	1028	918	855	658	474	301	186	253	384	622	815	915
	CDD10.0	157	0	0	0	0	2	21	87	43	4	0	0	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	38	0	0	0	0	0	3	29	6	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	2.9	2.7	2.8	3.4	3.1	3.3	3.5	2.9	2.6	2.8	2.7	2.4	2.8
Precipitation	PrecAvg	444	25	23	22	18	28	51	78	60	42	39	28	26
	PrecMax	582	47	57	48	45	95	102	202	107	93	78	58	80
	PrecMin	312	5	5	4	0	0	0	25	22	9	12	3	5
	PrecStd	73	10	16	14	13	21	26	36	21	23	20	15	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	4.5	4.8	10.0	19.2	22.2	26.1	24.0	16.8	10.3	5.5	5.0
		MCWB	1.9	2.7	1.9	5.9	10.8	13.8	16.4	15.8	11.4	7.6	3.9	3.0
	2%	DB	1.5	1.6	3.1	6.6	14.9	19.0	23.6	21.0	14.5	8.3	3.5	3.3
		MCWB	-0.1	-0.1	1.0	3.5	9.0	12.0	15.5	14.9	10.7	6.3	2.3	1.6
	5%	DB	-1.1	-0.4	1.4	4.7	11.6	16.5	21.5	18.3	12.8	6.9	1.4	1.1
		MCWB	-2.2	-1.7	-0.3	2.0	6.8	10.8	14.7	13.7	9.9	5.2	0.4	-0.3
	10%	DB	-3.1	-2.5	-0.3	3.4	9.4	14.6	19.2	16.4	11.3	5.2	-0.1	-0.5
		MCWB	-4.0	-3.5	-1.8	1.1	5.8	9.9	14.0	12.6	9.0	3.8	-0.8	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.8	2.4	5.9	11.8	16.2	17.9	17.3	12.7	8.1	4.1	3.2
		MCDB	3.7	4.6	4.3	9.9	18.4	20.9	23.4	22.2	15.0	9.7	5.2	4.8
	2%	WB	0.1	0.0	1.1	3.8	9.1	13.5	16.6	15.6	11.5	6.5	2.4	1.7
		MCDB	1.4	1.5	2.9	6.4	13.9	16.7	21.6	19.1	13.7	8.2	3.4	3.3
	5%	WB	-2.1	-1.5	-0.3	2.2	7.3	11.8	15.5	14.4	10.4	5.3	0.6	-0.1
		MCDB	-1.1	-0.3	1.3	4.3	11.1	15.4	20.0	17.6	12.1	6.8	1.3	1.0
	10%	WB	-3.9	-3.4	-1.7	1.2	6.0	10.5	14.5	13.1	9.4	3.9	-0.7	-1.5
		MCDB	-3.0	-2.5	-0.4	3.1	9.1	13.8	18.3	15.8	11.0	5.1	0.0	-0.5
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	10.7	9.7	7.7	8.1	9.4	9.3	7.7	6.5	8.7	9.5
		MCDBR	9.5	9.6	9.0	9.4	12.6	12.2	13.9	13.3	10.4	7.4	8.1	7.4
		MCWBR	8.7	8.5	7.9	7.0	7.5	6.4	6.8	7.7	6.9	5.6	7.3	6.5
	5% WB	MCDBR	9.4	9.4	8.5	8.5	11.9	10.4	11.7	11.8	8.5	6.8	7.6	7.2
		MCWBR	8.8	8.6	7.6	6.4	7.3	6.0	6.3	7.1	6.0	5.5	7.0	6.5
Clear-Sky Solar Irradiance	taub	0.177	0.233	0.275	0.302	0.321	0.326	0.332	0.318	0.287	0.248	0.179	N/A	
	taud	1.916	2.090	2.183	2.215	2.359	2.490	2.494	2.511	2.559	2.270	1.964	N/A	
	Ebn at Noon	74	551	752	840	869	876	857	830	771	540	62	0	
	Edh at Noon	6	44	76	100	99	90	87	76	55	36	5	0	
All-Sky Solar Radiation	RadAvg	0.02	0.44	1.92	3.91	5.06	4.94	4.59	3.27	1.76	0.56	0.08	0.00	
	RadStd	0.00	0.04	0.08	0.17	0.43	0.38	0.41	0.25	0.18	0.07	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page